

Final Portfolio Project

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Introduction

Sleep is one of the first things students sacrifice when school gets busy, but it plays a major role in stress, focus, and academic performance. Late-night studying and inconsistent routines often create a “night-owl” pattern that feels normal - but might carry hidden costs. In this project, we explore how sleep duration relates to sleep quality, stress levels, study habits, and academic outcomes. To do this, we use three real datasets: the Sleep Health and Lifestyle dataset, which includes sleep duration, sleep quality, stress, age, and gender; the Student Performance dataset, which contains performance index, previous scores, study time, and reported sleep hours; and the Student Performance Factors dataset, which adds information on exam scores, study hours, and additional behavioral variables.

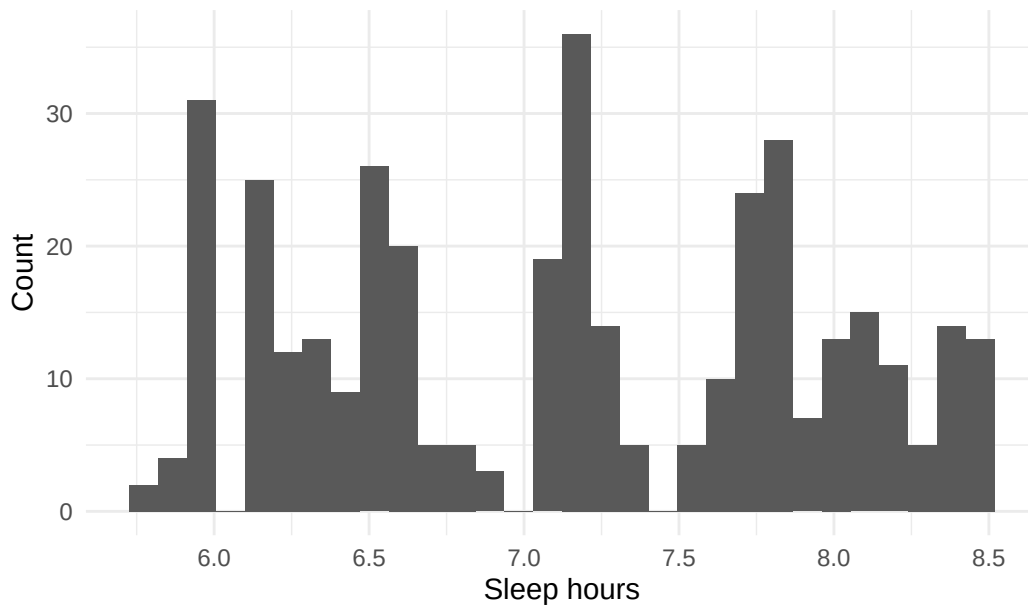
By analyzing these datasets individually and together, we look for consistent patterns connecting sleep to academic performance and well-being. Our visualizations, summaries, and interactive Shiny app help us ask whether students who sleep more report lower stress, whether sleep quality improves with longer sleep, and whether better sleep is linked to stronger academic outcomes. Although sleep is not the only factor shaping performance, our results suggest that adequate sleep is associated with better well-being and slightly higher academic achievement. This helped as highlight the importance of healthy sleep habits in a student’s daily life.

Research Questions

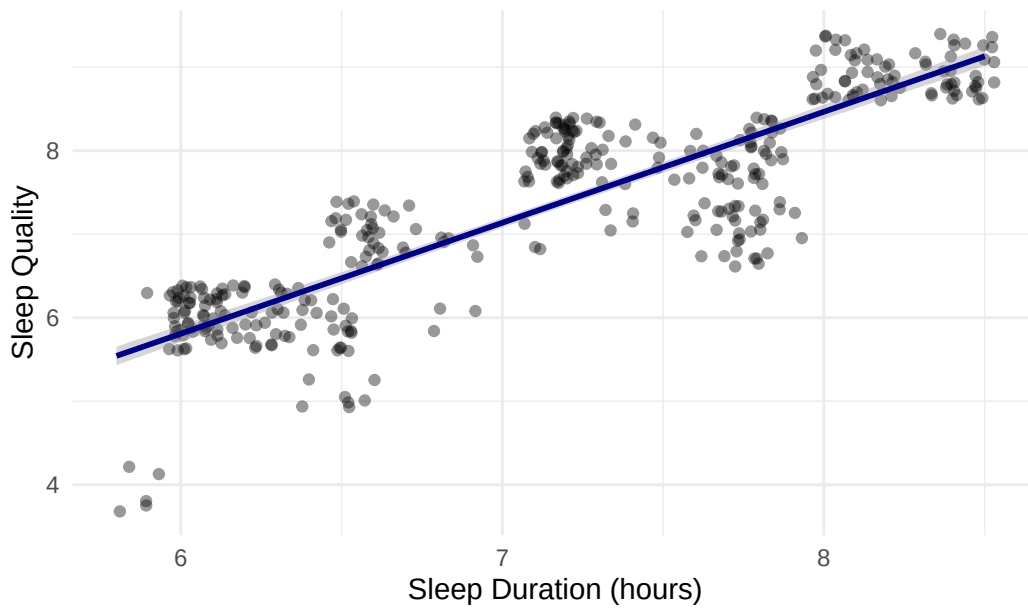
We focused on how sleeping relates to stress and academic performance using three datasets on sleep and student outcomes. Our project talks about three questions. The first one was: What do general sleep patterns look like in the Sleep Health and Lifestyle data? Which entails sub-questions such as How much do people sleep on average? How are sleep duration, sleep quality, and stress levels related? The second question was: How are sleep and academic performance related in student-level data? Which also covers sub-questions like, Is more sleep associated with higher performance index and exam scores? How does sleep compare to study time in predicting performance? And the last question was: How do sleep and study behaviors interact? Which also had sub-questions like Do students who sleep more also study more (or are they more efficient)? Are there “trade-offs” between sleep, stress, and performance?

Interpreting Sleep Patterns

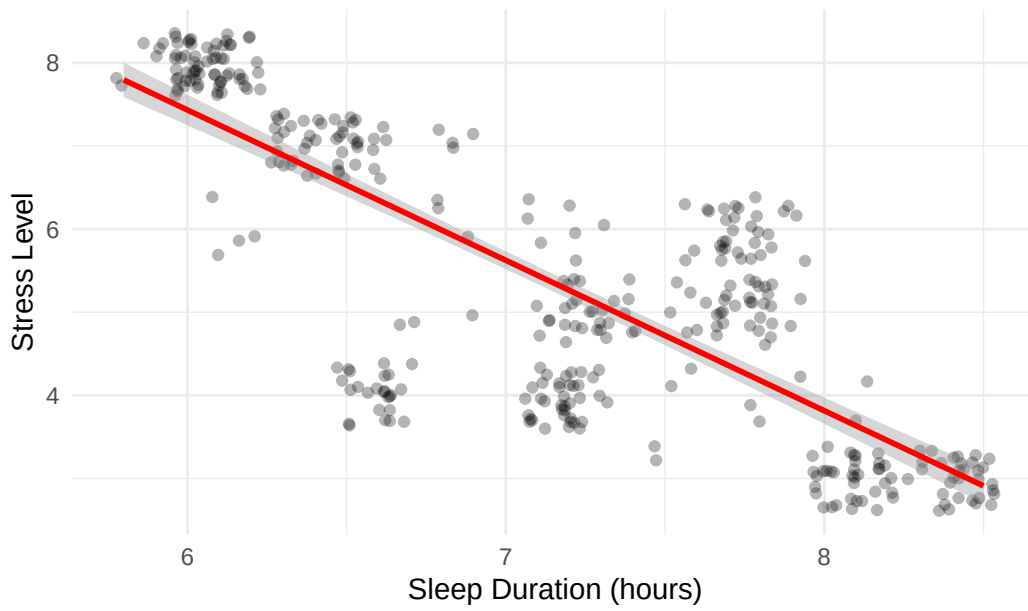
Distribution of Sleep Duration



Sleep Duration vs Sleep Quality

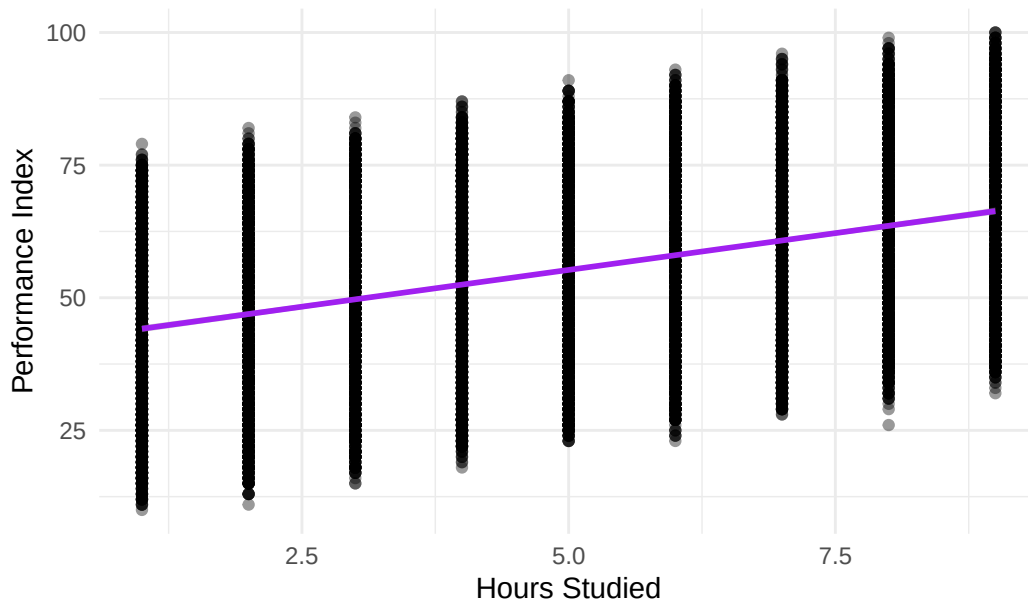


Sleep Duration vs Stress Level

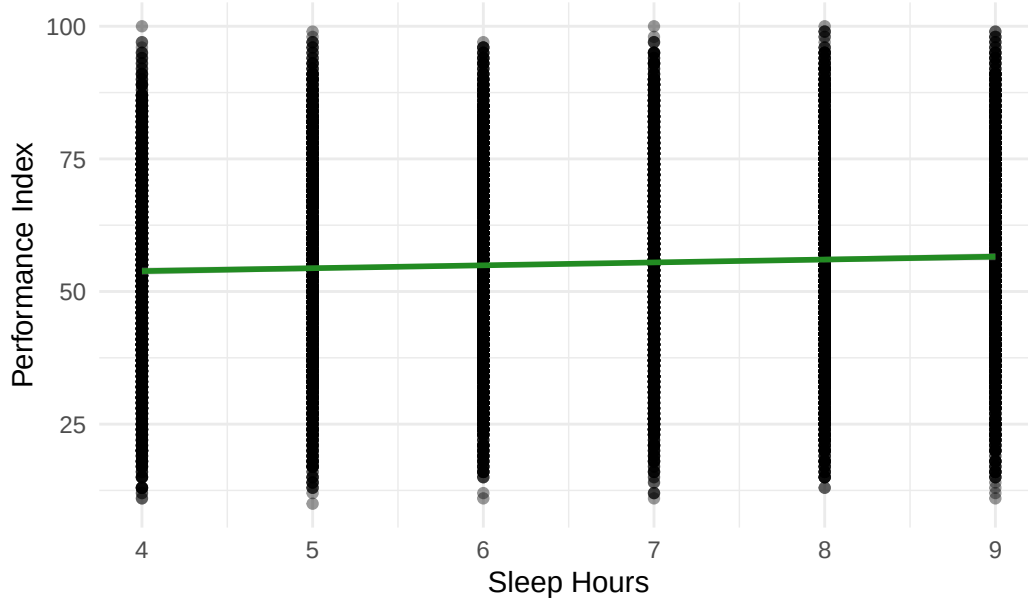


Sleep, Study Time, and Performance (Student Dataset)

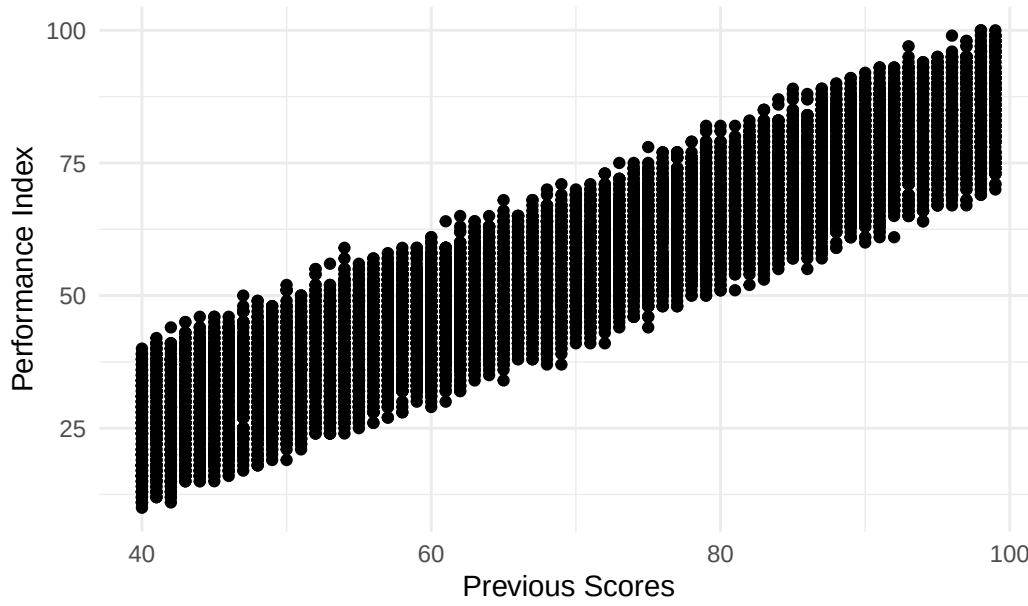
Study Time vs Performance Index



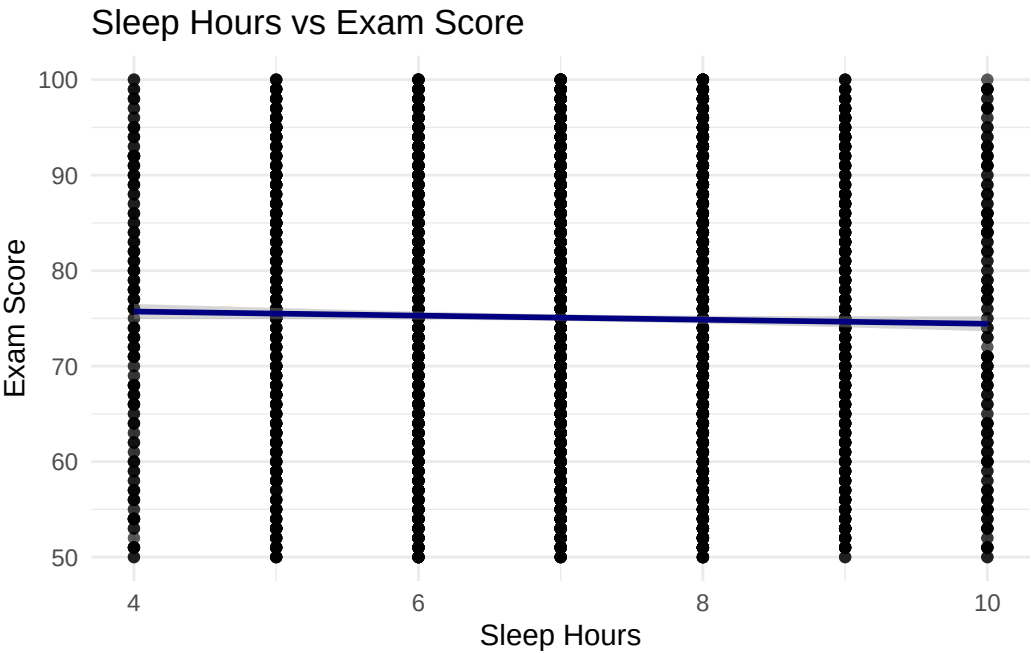
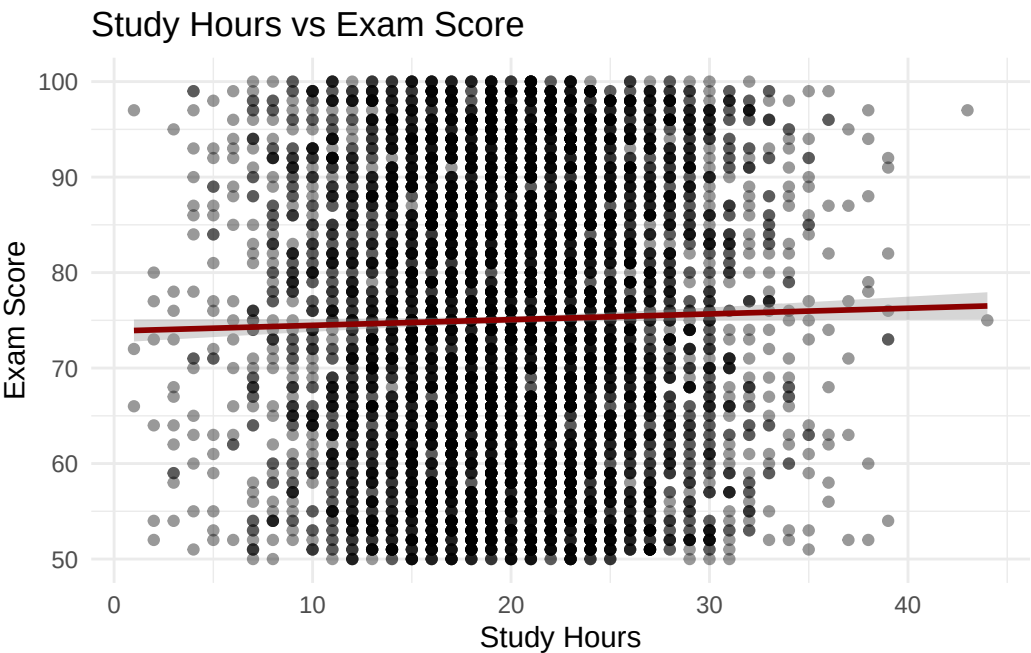
Sleep Hours vs Academic Performance



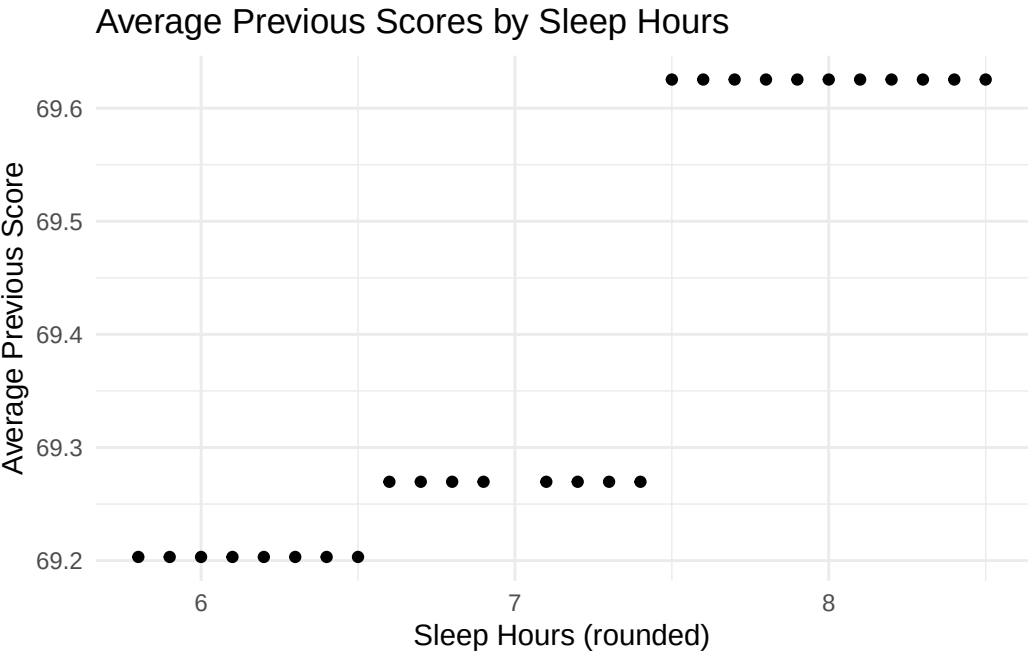
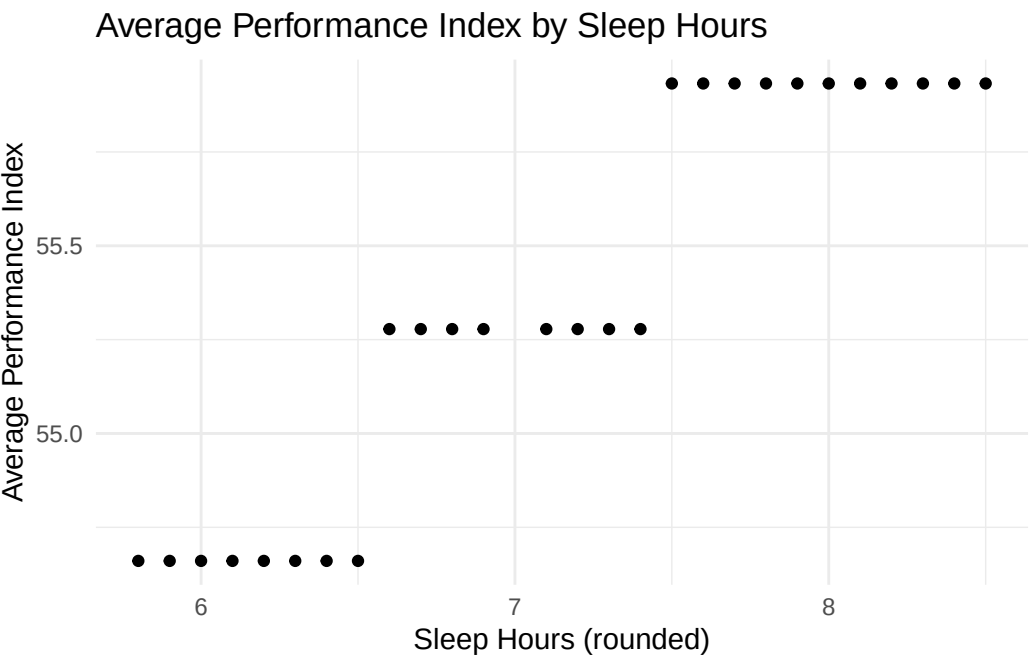
Previous Scores vs Academic Performance



Study Behavior and Exam Scores (Factors Dataset)



How Sleep Connects to Performance Across Datasets

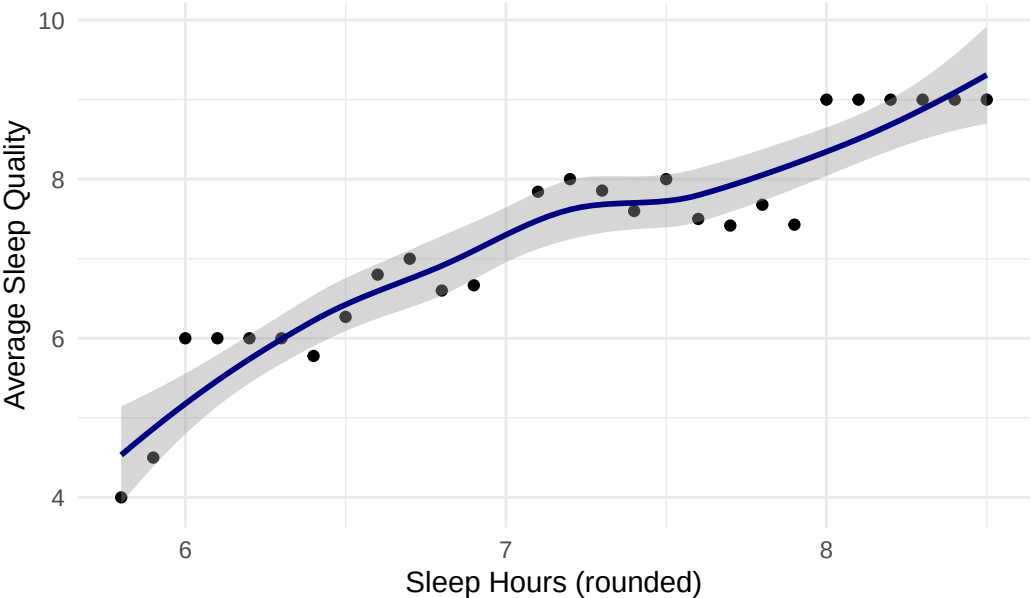


A dot plot showing the number of books read by students. The horizontal axis is labeled with 6, 7, and 8. There are 8 dots above 6, 4 dots above 7, and 10 dots above 8.

Average Exam Score by Sleep Hours



Average Sleep Quality by Sleep Hours



Average Stress Level by Sleep Hours

